

American Sign Language Recognition

TAA 2nd Assignment

Rodrigo Abreu, 113626

Eduardo Lopes, 103070

João Neto, 113482



A



B



C



D



E



F



G



H



I



K



L



M



N



O



P



Q



R



S



T



U



V



W



X



Y

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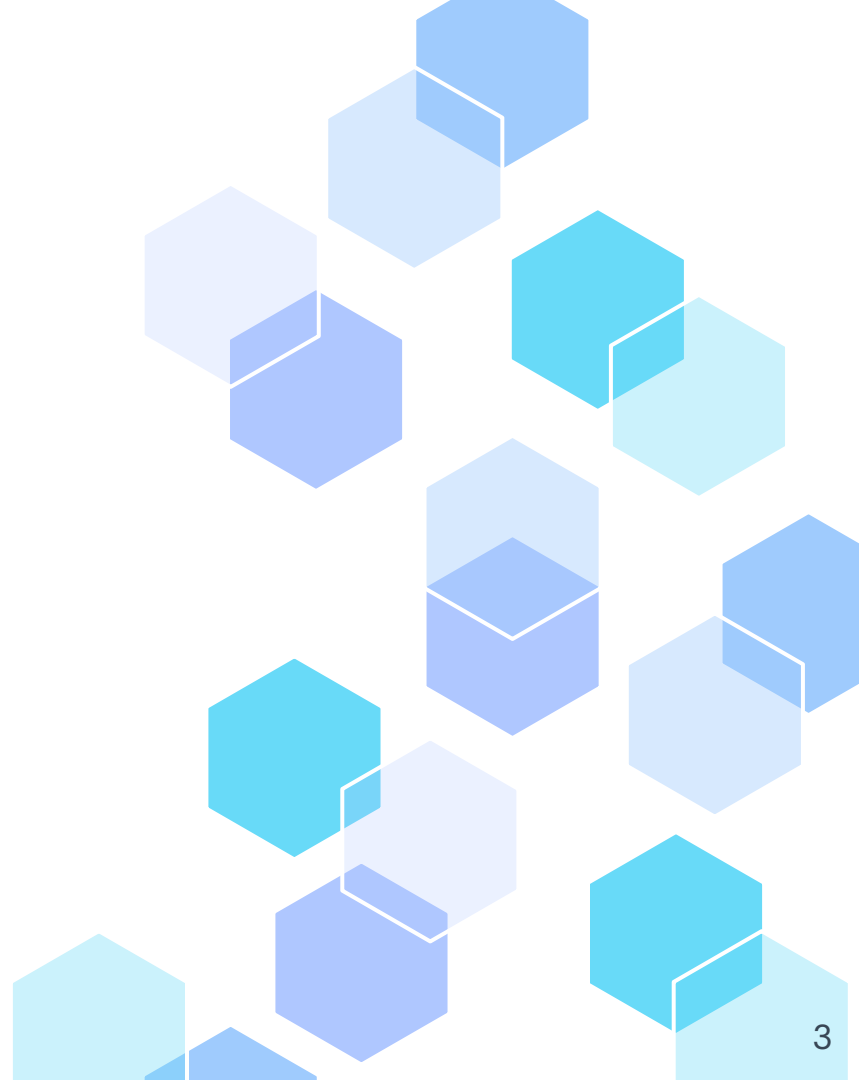
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01

Introduction

Problem description and dataset



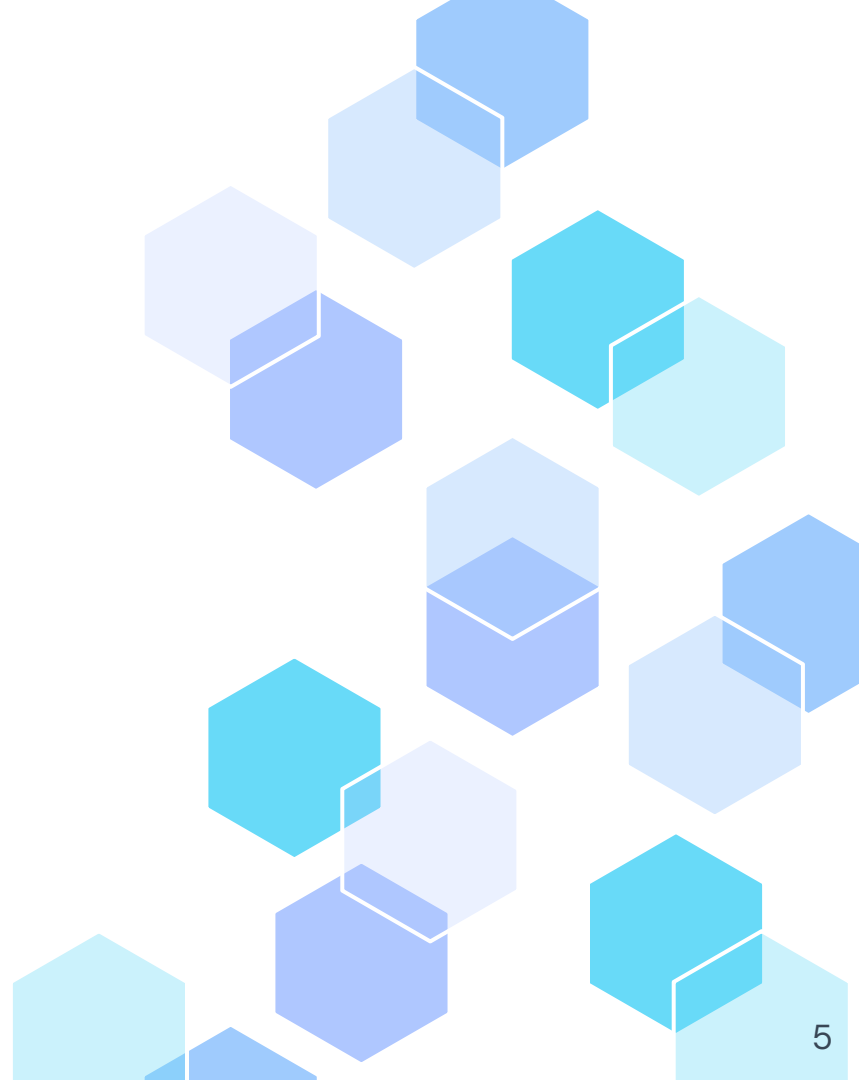
Introduction

- Communication gap between deaf/hard-of-hearing individuals and non-signers
- Need for automatic recognition of American Sign Language (ASL) letters
- Develop a system to recognize static ASL fingerspelling (24 letters) using machine learning
- The results provide insights into the trade-offs between model complexity and classification performance

02

State of Art

Modern and advanced related work



State of Art

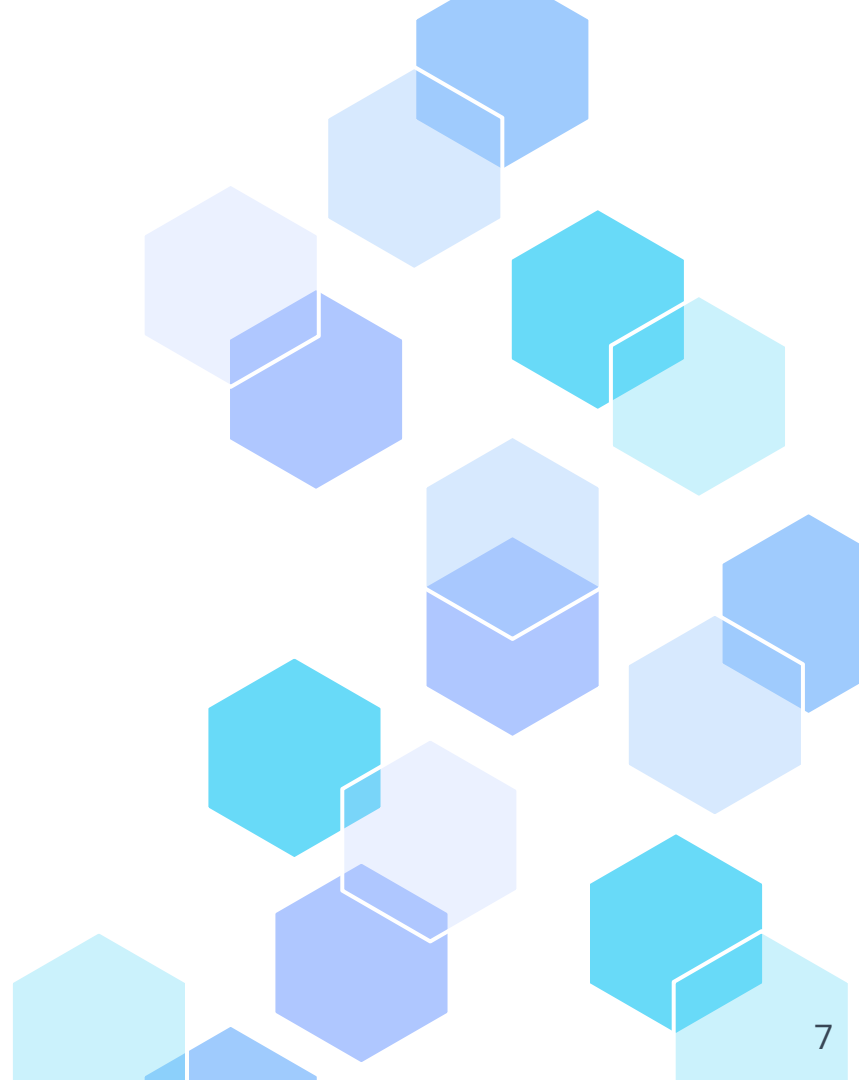
- CNNs dominate static ASL recognition due to spatial feature learning.
- Enhancements like synthetic augmentation, filtering, and dynamic modeling improve robustness
- Depth images boosted accuracy but need specialized hardware.
- Most models struggle with varying skin tones & lighting.



03

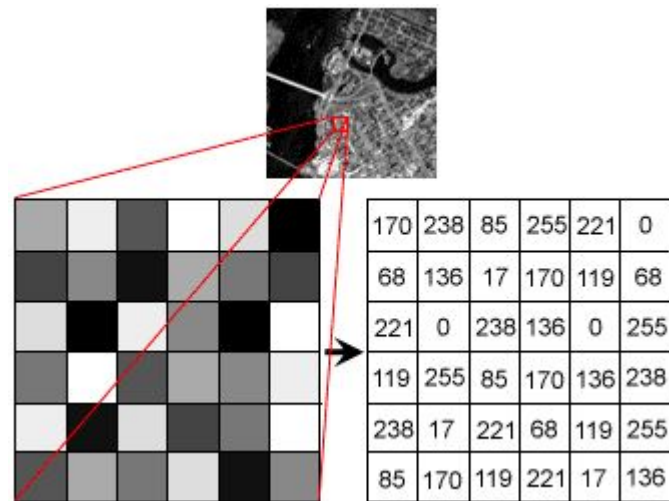
Data Analysis

Data description, visualization and statistics



3.1 Dataset

- **24 classes** of letters (excluding J and Z);
- 1 training file (27455 examples);
- 1 test file (7172 examples);
- Each example has **784 features**, 1 for each pixel value (28x28 image);
- Each value is number between 0 and 255. (grayscale pixel value).



3.2 Data Visualization

- Reconstruction of the some examples of the dataset from the pixel information;
- Some **similar signals** that may cause classification problems;
- Classes O (A), 4 (E), 11 (M), 12 (N), and 17 (S) are all represented by a closed hand with slight differences;
- Much **redundant Information** (background).



Example Image per Label

Label: 0 (A)



Label: 1 (B)



Label: 2 (C)



Label: 3 (D)



Label: 4 (E)



Label: 5 (F)



Label: 6 (G)



Label: 7 (H)



Label: 8 (I)



Label: 9 (K)



Label: 10 (L)



Label: 11 (M)



Label: 12 (N)



Label: 13 (O)



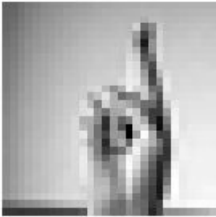
Label: 14 (P)



Label: 15 (Q)



Label: 16 (R)



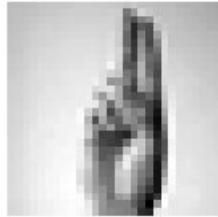
Label: 17 (S)



Label: 18 (T)



Label: 19 (U)



Label: 20 (V)



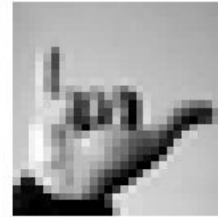
Label: 21 (W)



Label: 22 (X)

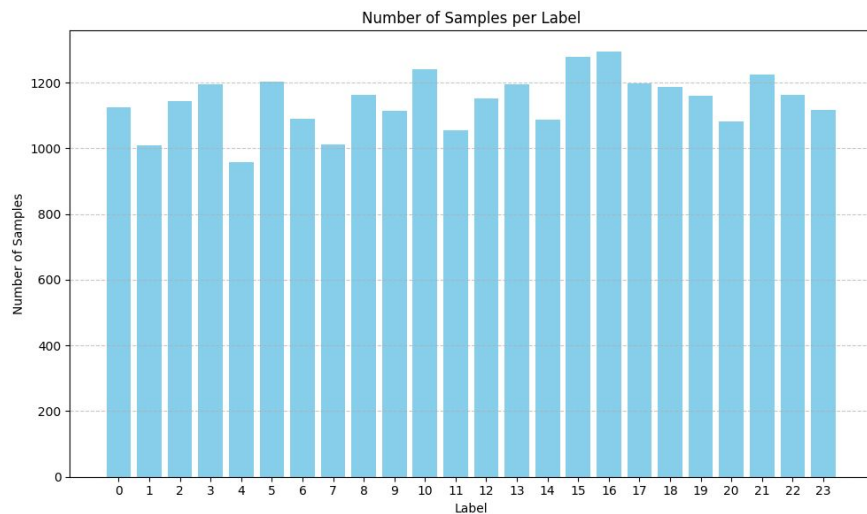


Label: 23 (Y)



3.3 Statistical Analysis

- We may have enough data to avoid overfitting: $27455/784 \approx 35$. (way ahead rule of thumb 10)
- Class examples distribution looks **balanced** enough.
- Balance techniques will not be necessary.



04

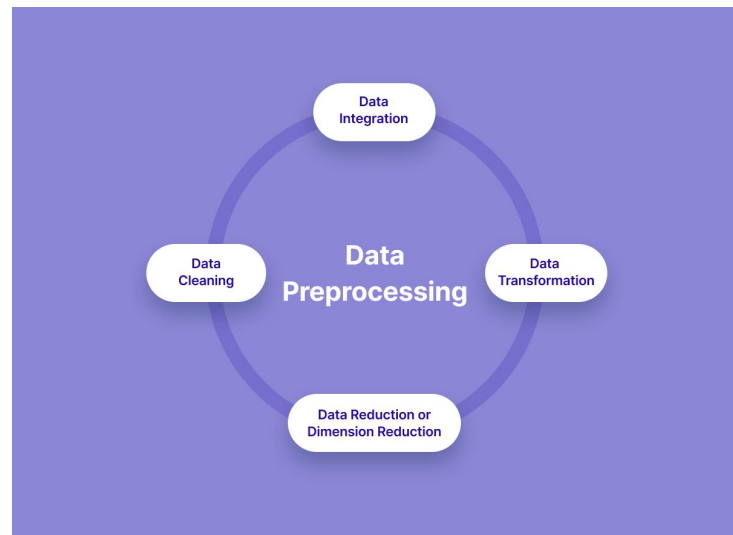
Data Preprocessing

Data preprocessing, for ML models usage

Data Preprocessing

Common steps for all:

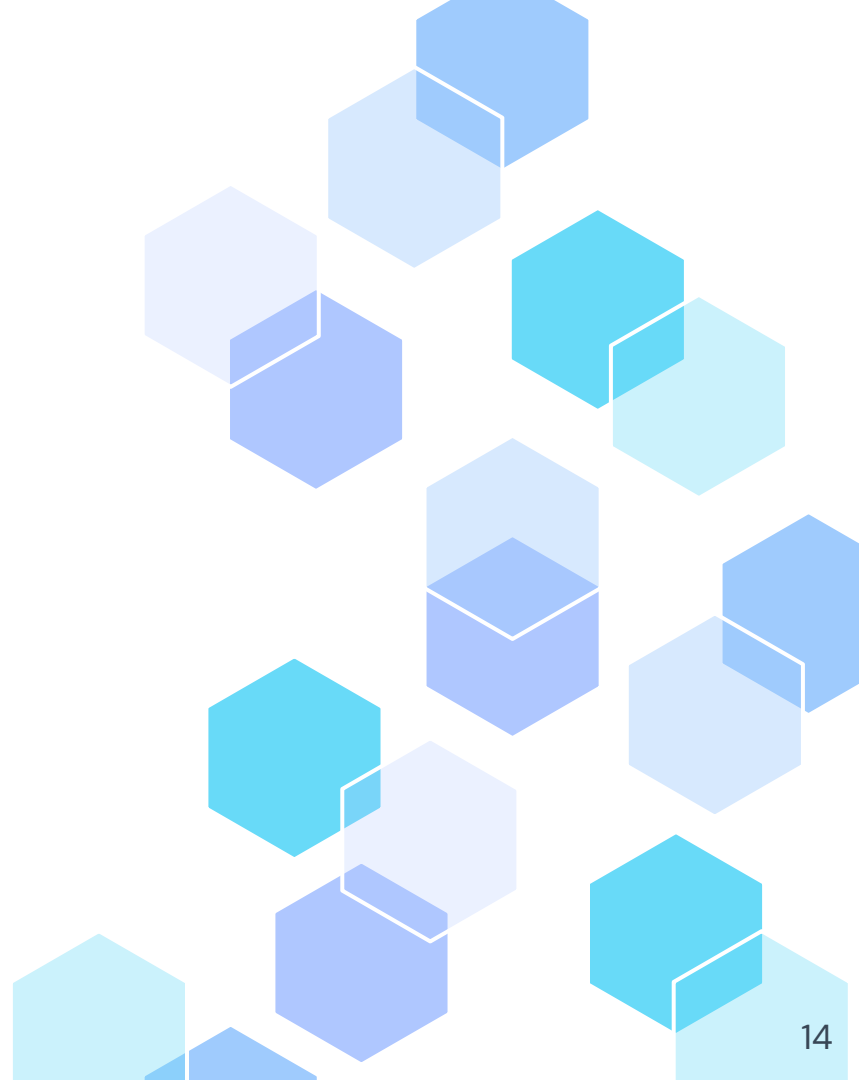
- Load train and test data for each variable;
- Remove headers;
- Separate label column;
- Apply **normalization**: divide train and test data pixel values by 255 (values in interval $[0, 1]$);
- This improves performance and convergence speed for ML models.



05

ML Algorithms

ML Algorithms implemented and results.



5.1

Softmax Logistic Regression



Softmax Logistic Regression

Generalization of the classic logistic regression for multiclass problems using the softmax function.

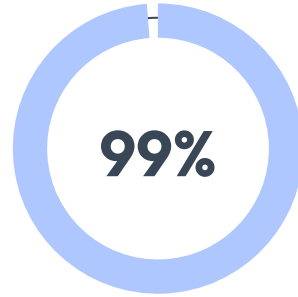
Why use this Algorithm?

- Multiclass classification.
- Even though limited for this implementation, is a great starting point.

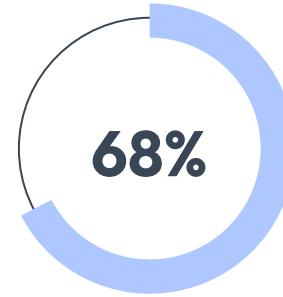
Steps applied:

- Data split: 80% training, 20% validation
- Hyperparameter tuning: $C = 4.6416$
- L2 Regularization: enhances generalization by penalizing large weights

Results



**Validation
Accuracy**



Test Accuracy

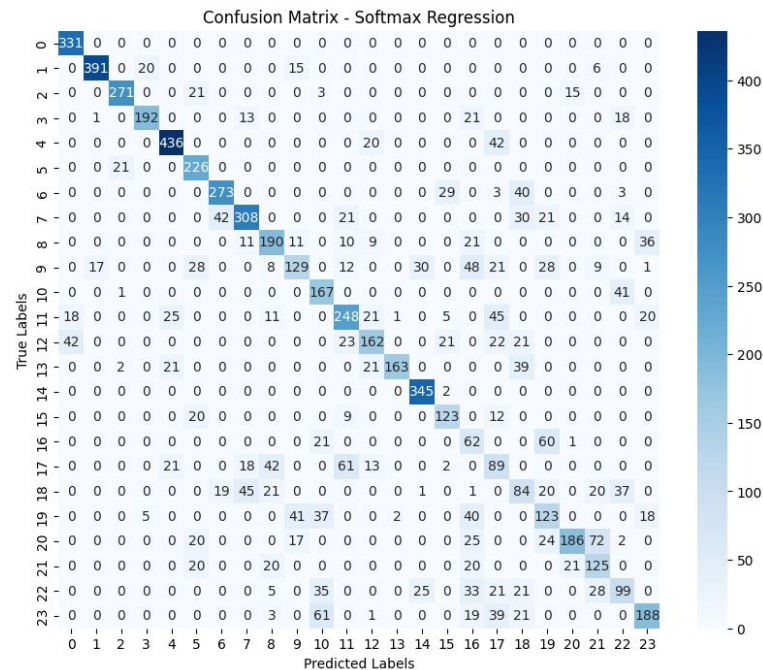
Big accuracy drop from validation to test results. SLR is limited for this type of task. ASL gesture images, exhibit complex, nonlinear relationships that this model is not equipped to capture.

Reports

Classification Report

	precision	recall	f1-score	support
0	0.85	1.00	0.92	331
1	0.96	0.91	0.93	432
2	0.92	0.87	0.90	310
3	0.88	0.78	0.83	245
4	0.87	0.88	0.87	498
5	0.67	0.91	0.78	247
6	0.82	0.78	0.80	348
7	0.78	0.71	0.74	436
8	0.63	0.66	0.65	288
9	0.61	0.39	0.47	331
10	0.52	0.80	0.63	209
11	0.65	0.63	0.64	394
12	0.66	0.56	0.60	291
13	0.98	0.66	0.79	246
14	0.86	0.99	0.92	347
15	0.68	0.75	0.71	164
16	0.21	0.43	0.29	144
17	0.30	0.36	0.33	246
18	0.33	0.34	0.33	248
19	0.45	0.46	0.45	266
20	0.83	0.54	0.65	346
21	0.48	0.61	0.54	206
22	0.46	0.37	0.41	267
23	0.71	0.57	0.63	332
accuracy			0.68	7172
macro avg	0.67	0.67	0.66	7172
weighted avg	0.71	0.68	0.69	7172

Confusion Matrix



Most Confused Signals

Label: 20 (V)



Label: 21 (W)



Label: 17 (S)



Label: 11 (M)



Label: 10 (L)

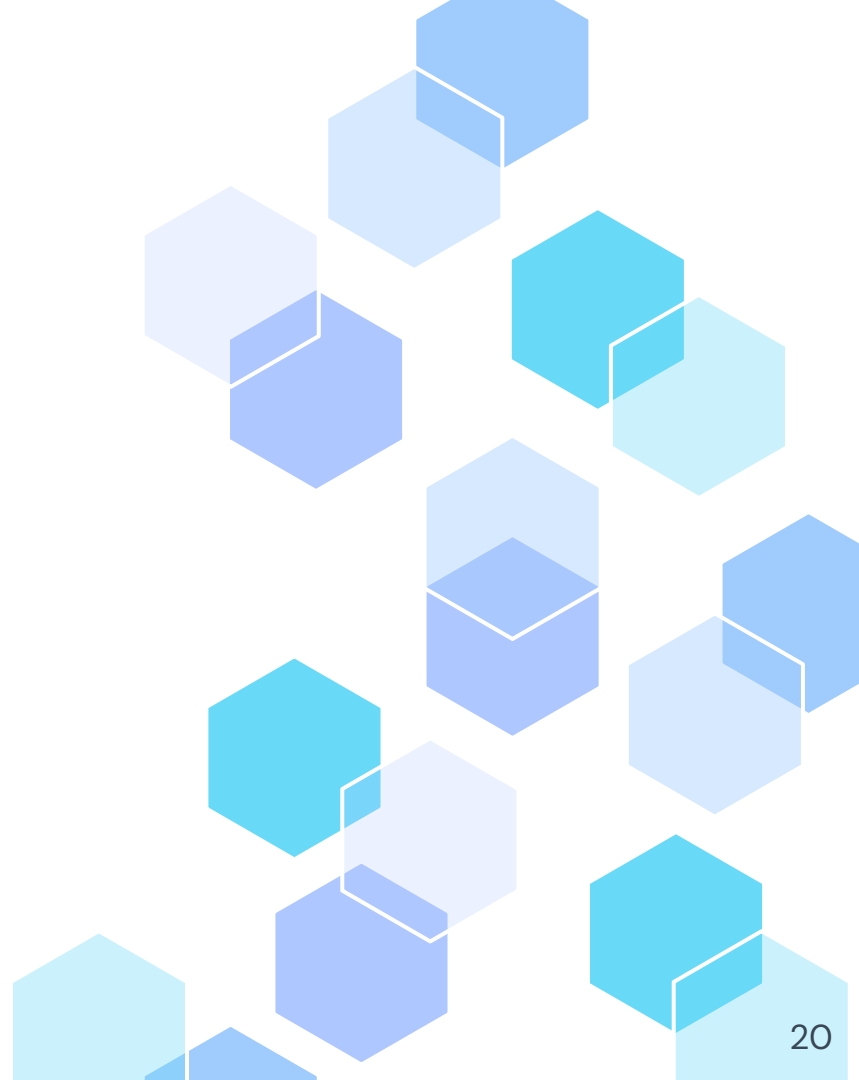


Label: 23 (Y)



5.2

One-vs-All Multiclass Classification with Logistic Regression



One-vs-All Multiclass Classification with Logistic Regression

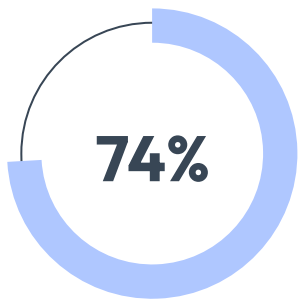
Why use this Algorithm?

- Easy to understand and use
- Works well for multiclass problems
- Good starting point before trying more complex models
- Can perform well with the right features

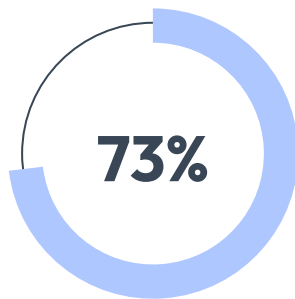
Hyperparameter tuning:

- Alpha: 0.1
- Lambda: 0.001

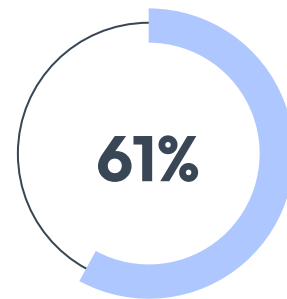
First development



**Training
Accuracy**



**Validation
Accuracy**



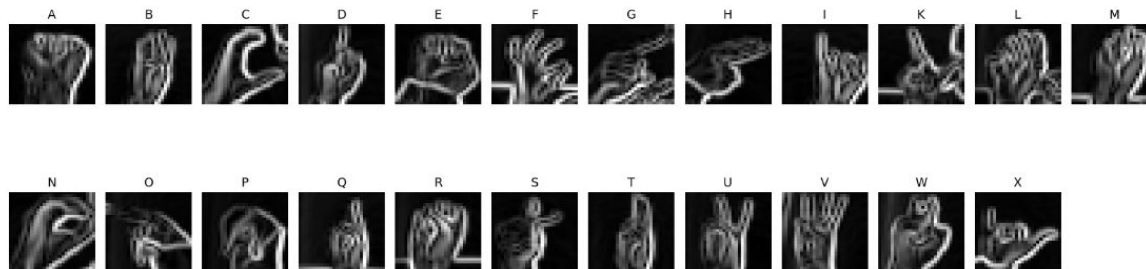
Test Accuracy

Using raw images resulted in moderate performance, with a noticeable gap between validation and test accuracy, suggesting the need for better feature extraction.

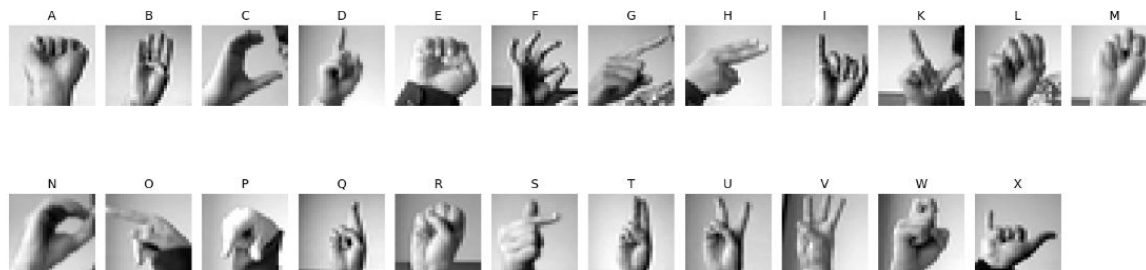
Feature Extraction Technique for Image Processing

Comparison of raw images and Sobel-processed images for signs A-Z.

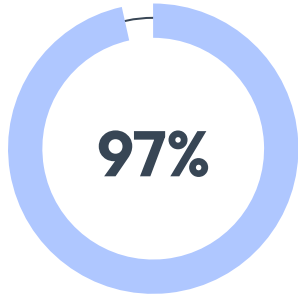
Sobel Edge Detection - 24 Letras (A-I, K-Y)



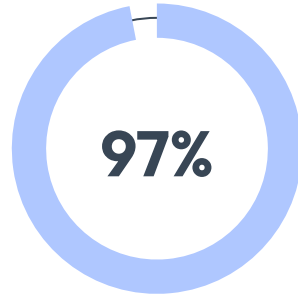
Imagens Originais - 24 Letras (A-I, K-Y)



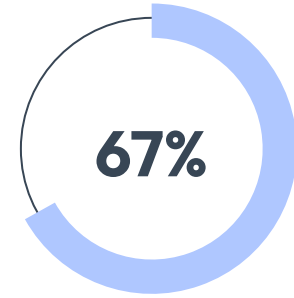
Final development - using a Edge Detection method for image preprocessing



**Training
Accuracy**



**Validation
Accuracy**



Test Accuracy

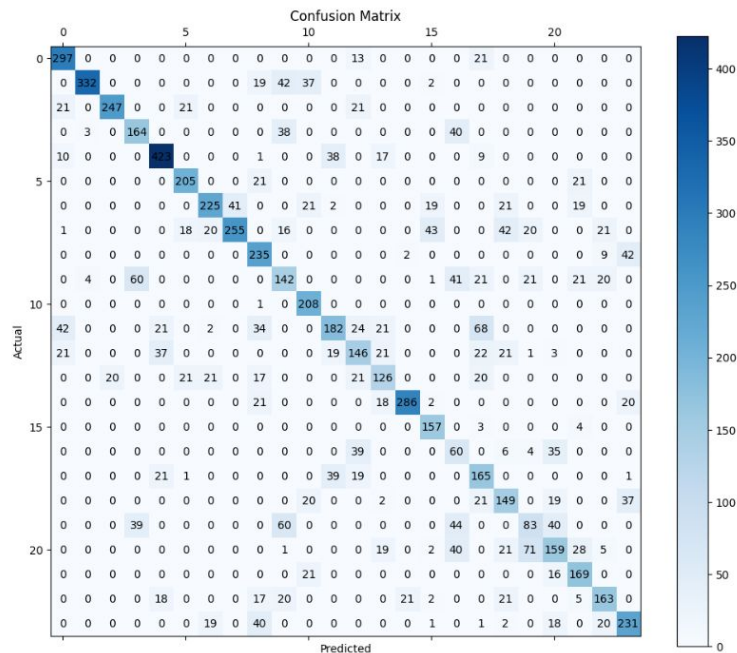
While edge detection improved training and validation accuracy to 97%, the lower test accuracy (67%) suggests some overfitting remains.

[illegible]

Classification Report

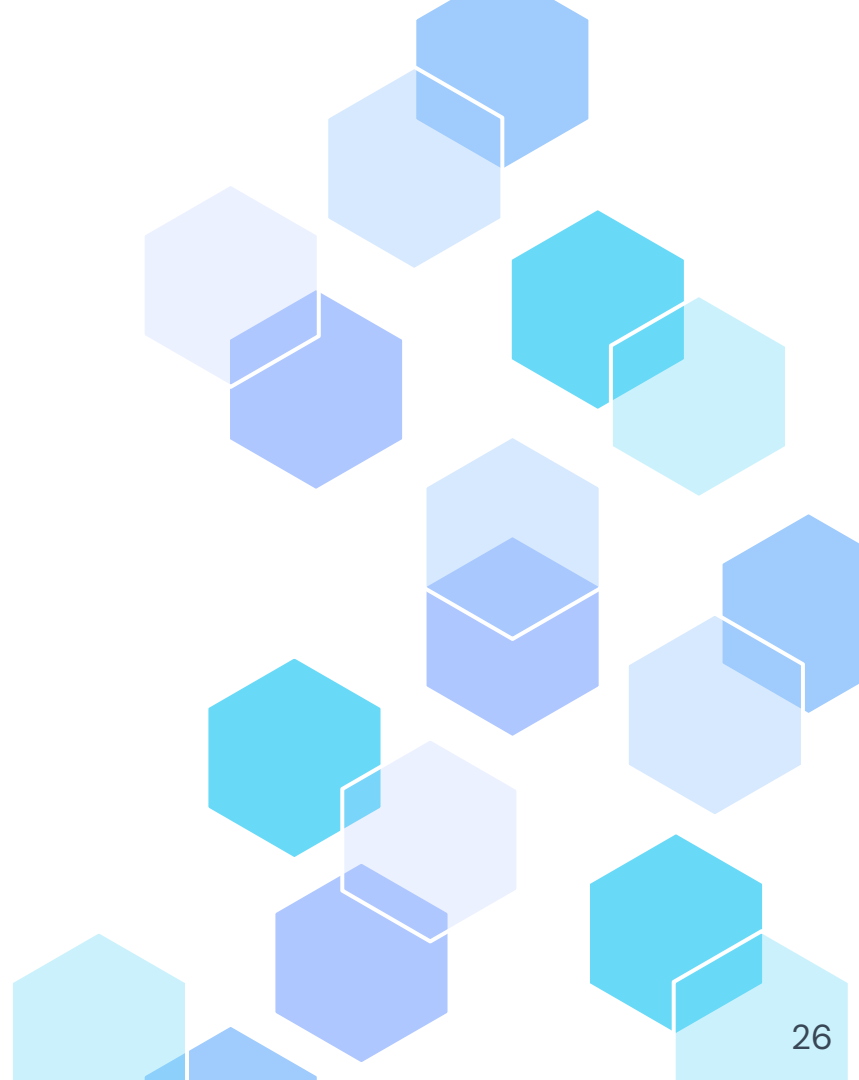
	precision	recall	f1-score	support
0	0.76	0.90	0.82	331
1	0.98	0.77	0.86	432
2	0.93	0.80	0.86	310
3	0.62	0.67	0.65	245
4	0.81	0.85	0.83	498
5	0.77	0.83	0.80	247
6	0.78	0.65	0.71	348
7	0.86	0.58	0.70	436
8	0.58	0.82	0.68	288
9	0.45	0.43	0.44	331
10	0.68	1.00	0.81	209
11	0.65	0.46	0.54	394
12	0.52	0.50	0.51	291
13	0.56	0.51	0.54	246
14	0.93	0.82	0.87	347
15	0.69	0.96	0.80	164
16	0.27	0.42	0.33	144
17	0.47	0.67	0.55	246
18	0.53	0.60	0.56	248
19	0.41	0.31	0.36	266
20	0.55	0.46	0.50	346
21	0.63	0.82	0.71	206
22	0.68	0.61	0.65	267
...				
accuracy			0.67	7172
macro avg	0.66	0.67	0.66	7172
weighted avg	0.69	0.67	0.67	7172

Confusion Matrix



5.3

K-Nearest Neighbors



K-Nearest Neighbors

KNN assigns class labels based on the majority vote of the K closest neighbors in the feature space.

Why use this Algorithm?

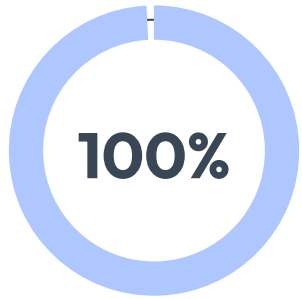
- Effective for image-based data
- Leverages the similarity of pixel patterns to distinguish between classes.

Steps applied:

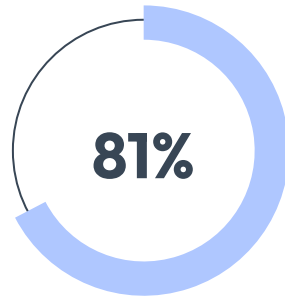
- Cross-validation to find the best K ($K=1$);
- PCA applied to reduce dimensionality by removing noise and redundant features;
- PCA with 95% of variance retained.

Results

K=1

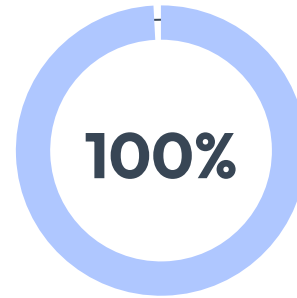


**Validation
Accuracy**

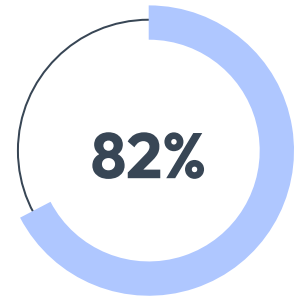


Test Accuracy

K=1 and PCA application



**Validation
Accuracy**



Test Accuracy

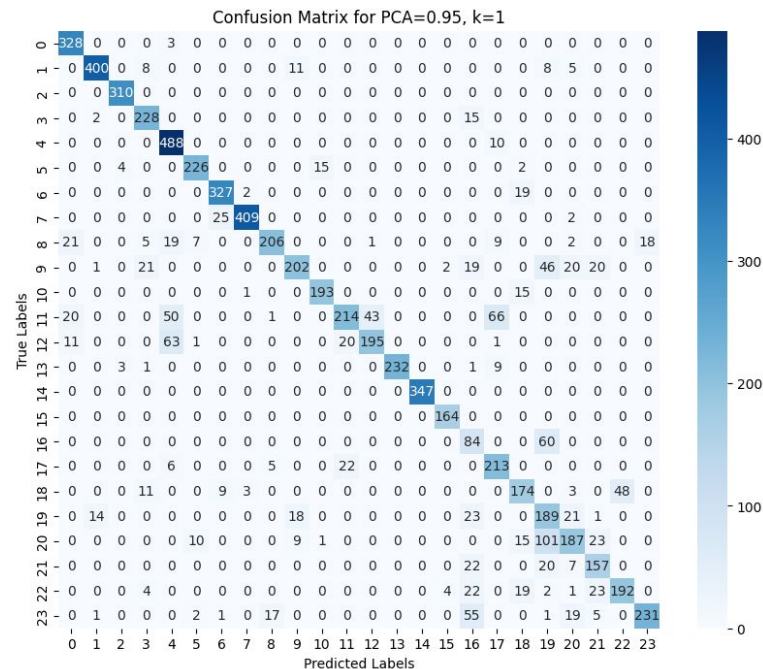
KNN implementation results on test accuracy were way higher than previous ones. PCA application only improved 1%, which was disappointing. This model shows signs of overfitting.

Reports

Classification Report

	precision	recall	f1-score	support
0	0.84	1.00	0.91	331
1	0.97	0.94	0.95	432
2	0.96	1.00	0.98	310
3	0.80	0.92	0.86	245
4	0.79	0.97	0.87	498
5	0.95	0.91	0.93	247
6	0.90	0.95	0.92	348
7	0.98	0.94	0.96	436
8	0.86	0.67	0.75	288
9	0.83	0.56	0.67	331
10	0.93	0.91	0.92	209
11	0.81	0.52	0.63	394
12	0.78	0.64	0.70	291
13	1.00	0.92	0.96	246
14	1.00	1.00	1.00	347
15	0.94	1.00	0.97	164
16	0.32	0.58	0.41	144
17	0.66	0.86	0.74	246
18	0.71	0.71	0.71	248
19	0.43	0.70	0.54	266
20	0.65	0.52	0.58	346
21	0.65	0.70	0.68	206
22	0.81	0.70	0.75	267
23	0.93	0.68	0.78	332
accuracy			0.81	7172
macro avg	0.81	0.80	0.80	7172
weighted avg	0.83	0.81	0.81	7172

Confusion Matrix



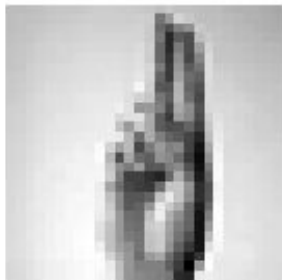
Low Precision Signals

Label: 16 (R)



35% Precision

Label: 19 (U)



44% Precision

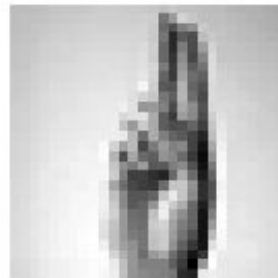
Most Confused Signals

Label: 20 (V)



Confused
as

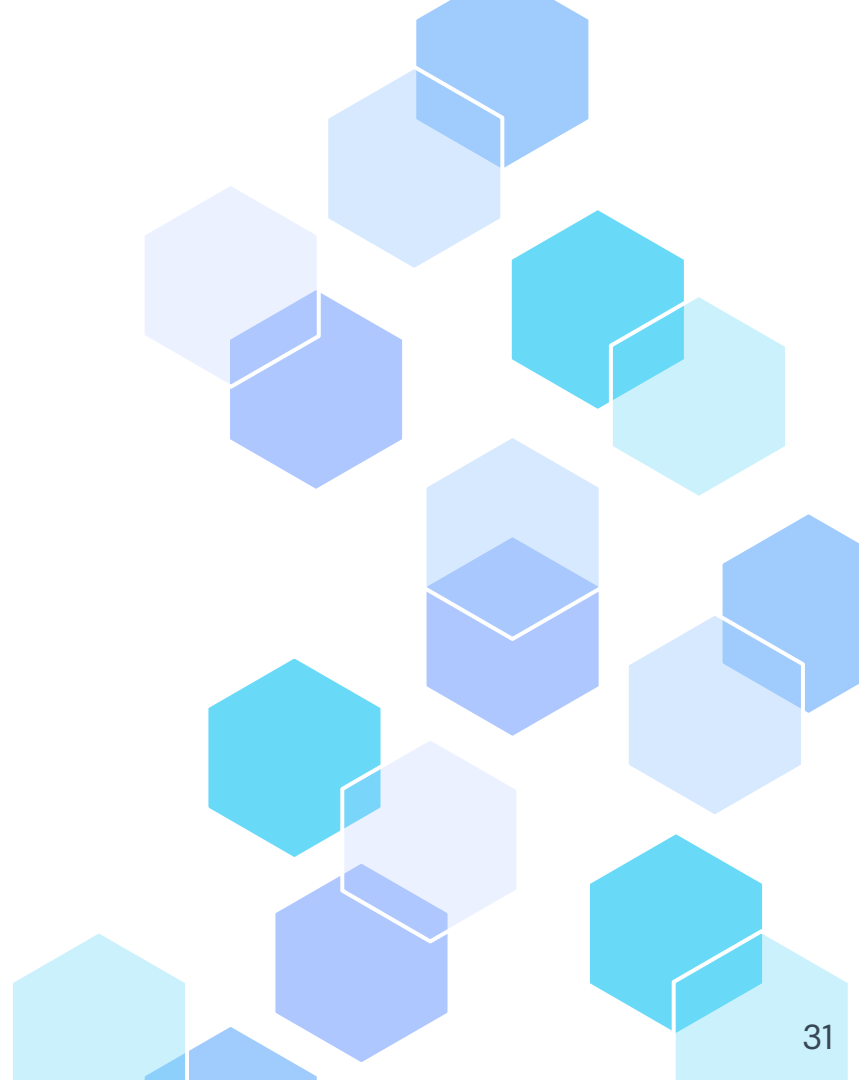
Label: 19 (U)



Confused 101 times

5.4

Support Vector Machine



Support Vector Machine (SVM)

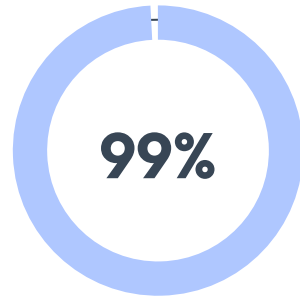
Why use this Algorithm?

- Effective in high-dimensional spaces
- Works well for clear margin of separation
- Robust to overfitting
- Can use kernel tricks for non-linear classification

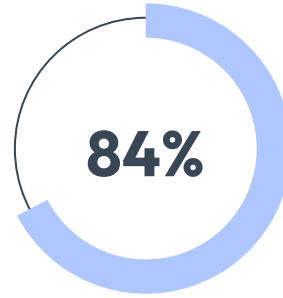
Hyperparameter tuning:

- C (Regularization parameter): 100
- Gamma: 0.035
- Kernel: rbf

Results



**Validation
Accuracy**



Test Accuracy

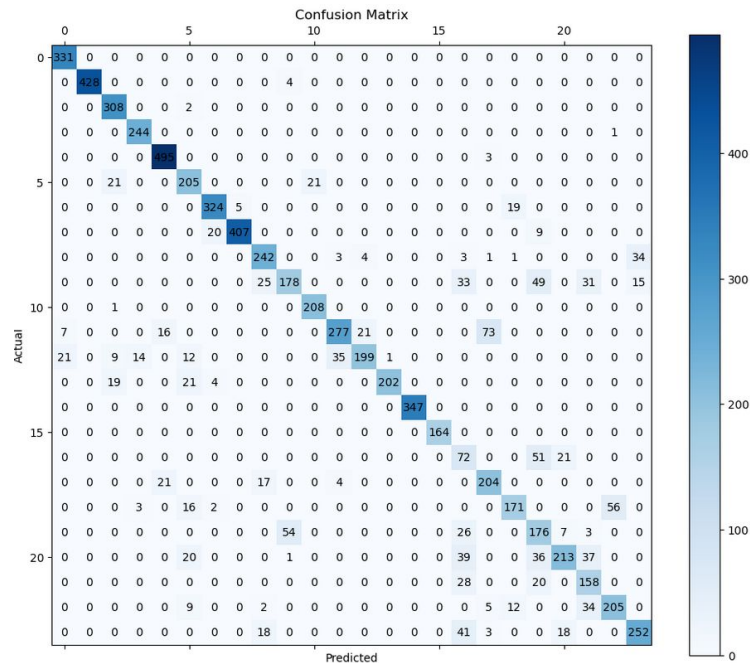
This discrepancy suggests a certain degree of overfitting, where the model generalizes less effectively to unseen data.

Reports

Classification Report

	precision	recall	f1-score	support
0	0.92	1.00	0.96	331
1	1.00	0.99	1.00	432
2	0.86	0.99	0.92	310
3	0.93	1.00	0.96	245
4	0.93	0.99	0.96	498
5	0.72	0.83	0.77	247
6	0.93	0.93	0.93	348
7	0.99	0.93	0.96	436
8	0.80	0.84	0.82	288
9	0.75	0.54	0.63	331
10	0.91	1.00	0.95	209
11	0.87	0.70	0.78	394
12	0.89	0.68	0.77	291
13	1.00	0.82	0.90	246
14	1.00	1.00	1.00	347
15	1.00	1.00	1.00	164
16	0.30	0.50	0.37	144
17	0.71	0.83	0.76	246
18	0.84	0.69	0.76	248
19	0.52	0.66	0.58	266
20	0.82	0.62	0.70	346
21	0.60	0.77	0.67	206
22	0.78	0.77	0.78	267
23	0.84	0.76	0.80	332
accuracy			0.84	7172
macro avg	0.83	0.83	0.82	7172
weighted avg	0.85	0.84	0.84	7172

Confusion Matrix



5.5

Convolutional Neural Networks



Convolutional Neural Networks using Keras

CNN automatically learns spatial hierarchies of features from raw images, making it ideal for visual recognition tasks.

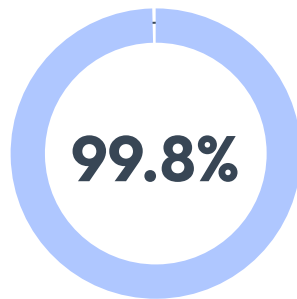
Why use this Algorithm?

- Try deep learning approach;
- The most suitable for this task.

Steps applied:

- Reshape the data from 1D to 3D;
- Artificially generate new training data, to avoid overfitting;
- Regularization techniques and learning rate adequation to improve generalization.

Results

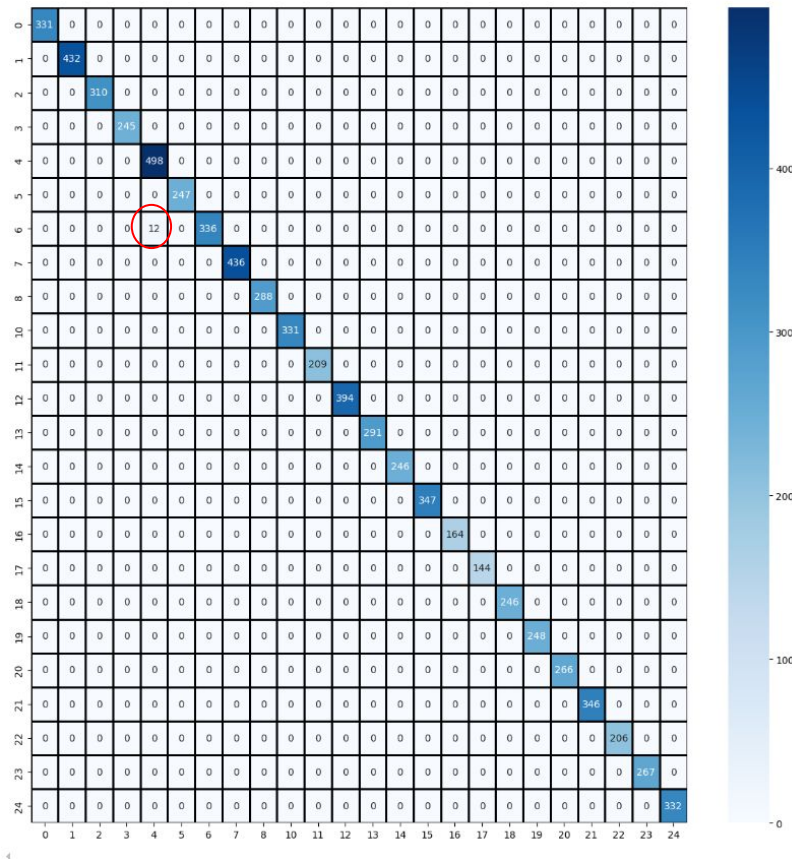


Test Accuracy

Impressive results, this model almost got a perfect accuracy score.

Confusion Matrix

The only limitation is a slight confusion of letter "G" with "E" (12 times).



Label: 6 (G)



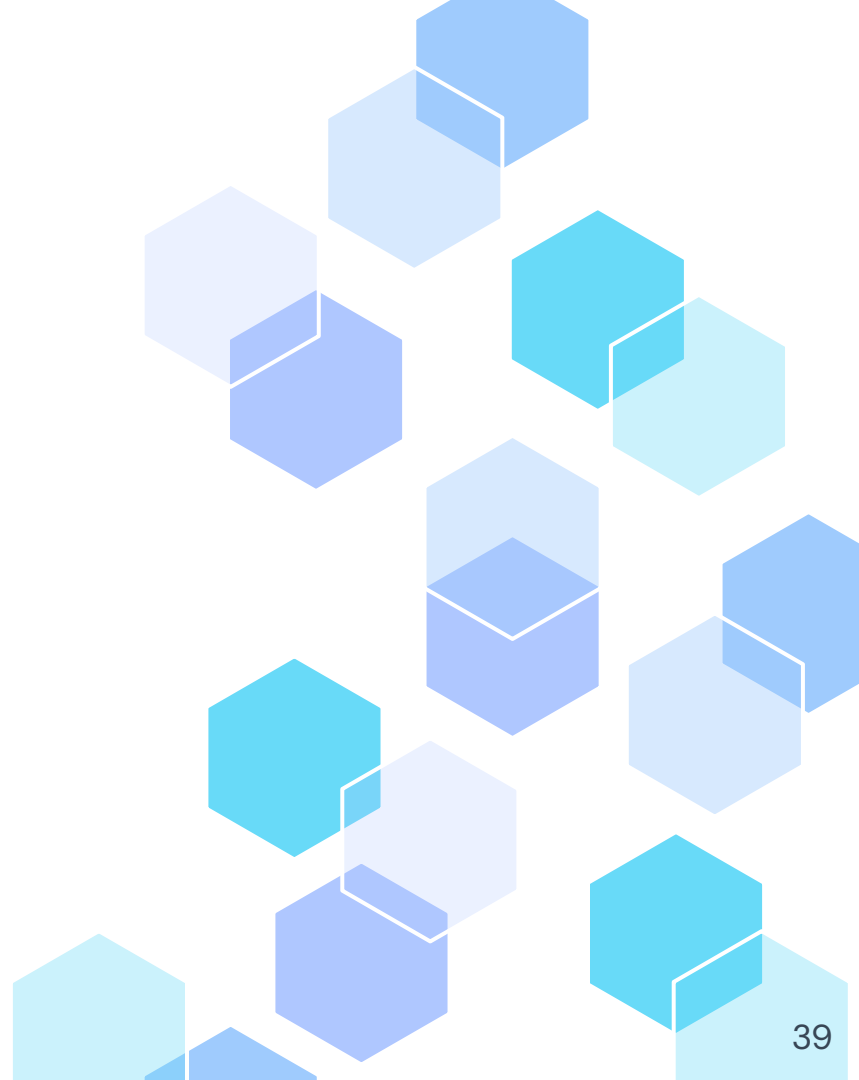
Label: 4 (E)



06

Conclusion

Overall results and final considerations



American Sign Language Recognition

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